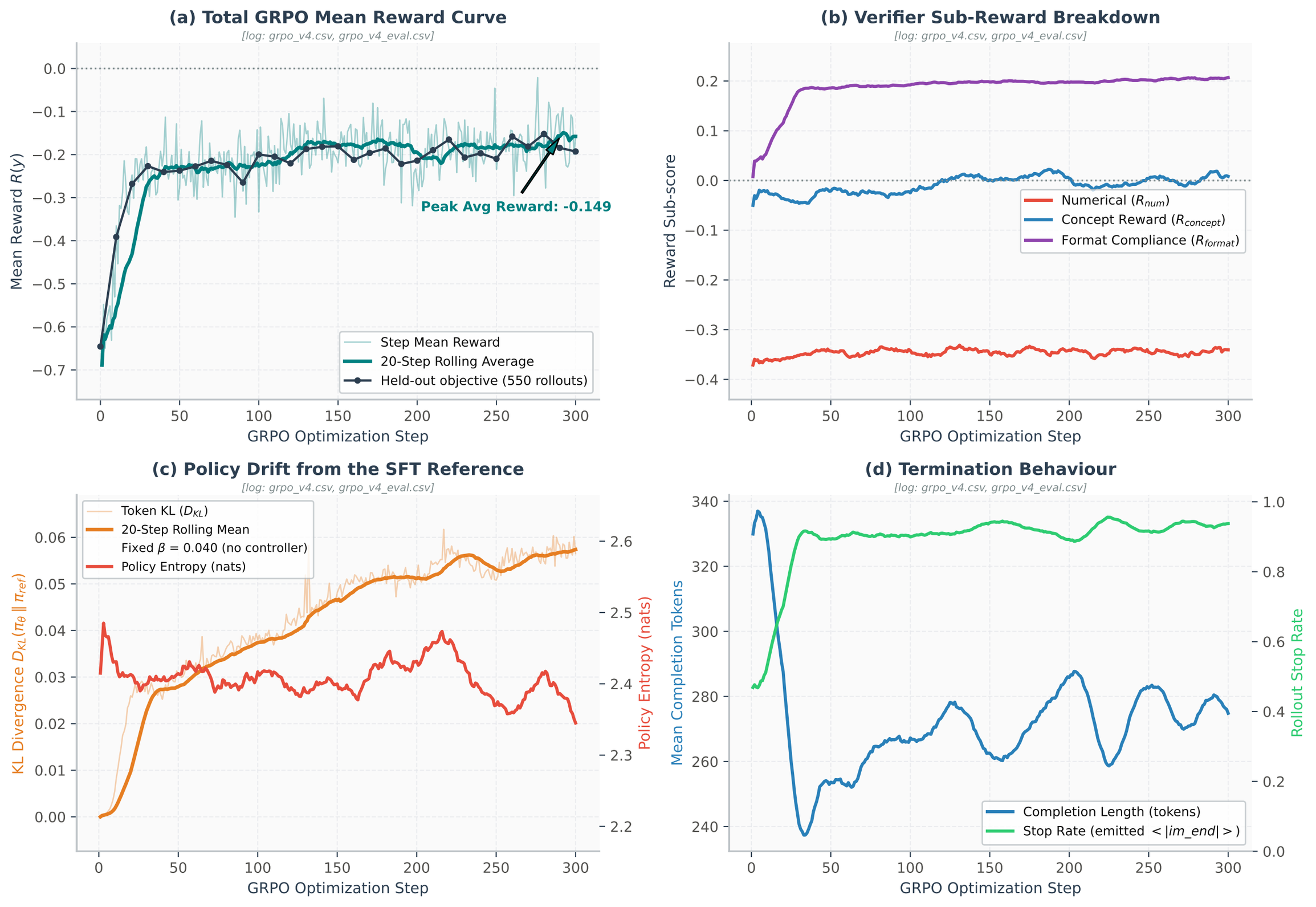

(a) Total GRPO Mean Reward Curve
[log: grpo_v4.csv, grpo_v4_eval.csv]
Mean Reward R(y)
GRPO Optimization Step
Peak Avg Reward: -0.149
Step Mean Reward
20-Step Rolling Average
Held-out objective (550 rollouts)
(b) Verifier Sub-Reward Breakdown
[log: grpo_v4.csv, grpo_v4_eval.csv]
Reward Sub-score
GRPO Optimization Step
Numerical (R_{num})
Concept Reward ($R_{concept}$)
Format Compliance (R_{format})
(c) Policy Drift from the SFT Reference
[log: grpo_v4.csv, grpo_v4_eval.csv]
KL Divergence $D_{KL}(\pi_\theta \| \pi_{ref})$
Policy Entropy (nats)
GRPO Optimization Step
Token KL (D_{KL})
20-Step Rolling Mean
Fixed β = 0.040 (no controller)
Policy Entropy (nats)
(d) Termination Behaviour
[log: grpo_v4.csv, grpo_v4_eval.csv]
Mean Completion Tokens
Rollout Stop Rate
GRPO Optimization Step
Completion Length (tokens)
Stop Rate (emitted <|im_end|>)